SESSION 2 — MORNING

Microeconomics

Supply, demand, firm costs, profit maximization and market structures

Session Roadmap — 2 Hours

Six sections. Microeconomics provides the toolkit for understanding firm behaviour, pricing power and industry profitability.

01 **Supply, Demand & Elasticity**
Demand curve • supply curve • equilibrium • PED • income elasticity • surplus

02 **Firm Costs & Revenue**
TR, MR, AR • TC, TFC, TVC, ATC, MC • short run vs long run • accounting vs economic profit

03 **Profit Maximisation & Shutdown**
MR=MC rule • WR International (CFA LM1) • London business shutdown (CFA LM1)

04 **Market Structures**
CFA LM1 Exhibit 13 table • perfect competition • monopolistic comp • oligopoly • monopoly

05 **Game Theory & Market Power**
Nash equilibrium: ArcCo/BatCo (CFA LM1) • CR • HHI: 5-firm market (CFA LM1)

06 **Practice & Takeaways**
5 worked problems • formula reference • CFA key examples review

Learning Objectives

- **Supply, Demand & Equilibrium**
  Apply the laws of supply and demand. Calculate and interpret price, income and cross-price elasticity. Understand consumer and producer surplus.
- **Firm Cost Analysis**
  Distinguish economic from accounting profit. Classify costs (TFC, TVC, ATC, MC). Describe short-run vs long-run cost curves and economies of scale.
- **Profit Maximisation & Shutdown**
  Apply MR=MC rule. Interpret WR International profit table (CFA LM1). Apply London business shutdown analysis (CFA LM1).
- **4 Market Structures**
  Compare perfect competition, monopolistic competition, oligopoly and monopoly using the CFA LM1 Exhibit 13 framework.
- **Game Theory & Nash Equilibrium**
  Interpret the ArcCo/BatCo Nash equilibrium payoff matrix (CFA LM1). Understand collusion incentives.
- **Measuring Market Power**
  Calculate 4-firm concentration ratio (86% from CFA LM1). Compute HHI for a 5-firm market (0.225 from CFA LM1). Interpret antitrust implications.

SECTION 01

Supply, Demand & Elasticity

The Demand Curve — Law of Demand

The demand curve shows the MAXIMUM quantity consumers will buy at each price. All else equal, higher price → lower quantity demanded.

Law of Demand: as price rises, quantity demanded falls (negative slope)

Factors that SHIFT the demand curve (change demand at all prices):

Shift RIGHT (demand increases)

- Income rises (for normal goods)
- Substitute goods become more expensive
- Complementary goods become cheaper
- Positive expectations about future prices
- Population increases

Shift LEFT (demand decreases)

- Income falls (for normal goods)
- Substitute goods become cheaper
- Complementary goods more expensive
- Negative economic expectations
- Tastes change away from the product

Supply Curve & Market Equilibrium

Law of Supply: as price rises, quantity supplied rises (positive slope)

At EQUILIBRIUM, market clears: quantity demanded = quantity supplied. Price adjusts to eliminate surplus or shortage.

Situation	Price vs Equilibrium	What Happens
Surplus (excess supply)	Price ABOVE equilibrium	Sellers cut price to sell unsold goods. Price falls until $Q_D = Q_S$
Shortage (excess demand)	Price BELOW equilibrium	Buyers bid up price. Price rises until $Q_D = Q_S$
Equilibrium	Price exactly right	No pressure to change. Market clears. No unsold inventory.

Shift in Supply: what raises/lowers the supply curve?

Supply shifts RIGHT (increases) when: input costs fall, technology improves, suppliers expect lower future prices, more producers enter market.

Supply shifts LEFT (decreases) when: input costs rise (e.g. energy price spike), taxes increase, natural disasters destroy capacity.

Price by itself causes a MOVEMENT along the curve — not a SHIFT of the curve.

Price Elasticity of Demand (PED) — Responsiveness of Demand

PED = % Δ Quantity Demanded / % Δ Price (always negative — report absolute value)

|PED|	Classification	What It Means	Examples
> 1	Elastic	Quantity changes MORE than price	Luxury goods, brand-switchable products, holidays
= 1	Unit elastic	Quantity changes SAME as price	Special case: total revenue unchanged
< 1	Inelastic	Quantity changes LESS than price	Petrol, insulin, tobacco, basic food
= 0	Perfectly inelastic	Quantity does NOT change with price	Life-saving medication, essential utilities
∞	Perfectly elastic	Quantity infinite at that price; zero at any higher price	Homogeneous goods in perfect competition

PED and Total Revenue

If |PED| > 1 (elastic): cutting price RAISES total revenue. Customers are responsive.

If |PED| < 1 (inelastic): cutting price REDUCES total revenue. Customers are unresponsive.

Investor insight: companies with inelastic demand have PRICING POWER — they can raise prices without losing significant volume (e.g. pharma, utilities, tech platforms with lock-in).

Income Elasticity & Cross-Price Elasticity

Income Elasticity = % Δ Q Demanded / % Δ Income **Cross-Price = % Δ Q_a / % Δ P^b**

These two elasticities tell us how demand for a product responds to changes outside the product's own price.

Type	Sign	Interpretation	Examples
Normal good	Income elasticity > 0	Demand rises when income rises. Rises faster if luxury (> 1).	Cars, foreign holidays, fine dining
Inferior good	Income elasticity < 0	Demand FALLS when income rises. Consumers switch to better alternatives.	Own-label food, bus travel, low-quality spirits
Luxury good	Income elasticity > 1	Demand rises FASTER than income. High income-sensitivity.	Designer goods, business-class travel, boats
Substitutes	Cross-price > 0	Higher price of B → more demand for A. Goods compete.	Pepsi/Coke, butter/margarine, Airbus/Boeing
Complements	Cross-price < 0	Higher price of B → LESS demand for A. Goods used together.	Cars/petrol, printers/ink, phones/cases

Investor application: understanding income elasticity helps forecast revenue through economic cycles. Luxury goods are highly cyclical (high income elasticity). Staples are defensive (low elasticity).

Consumer & Producer Surplus — Measuring Market Welfare

CS = willingness to pay – price paid PS = price received – minimum acceptable price

These two triangles represent the welfare gains from voluntary exchange. Maximum TOTAL surplus = maximum social welfare.

Consumer Surplus (CS)

Area between the demand curve and market price.
Buyers pay LESS than they were willing to pay.

Example: Consumer willing to pay £80 for a product that costs £50. CS = £30.

CS falls when: price rises, supply disrupted, monopoly charges higher prices.

Producer Surplus (PS)

Area between market price and the supply curve.
Sellers receive MORE than minimum they would have accepted.

Example: Producer willing to accept £30 but receives £50. PS = £20.

PS rises when: prices rise, production efficiency improves.

Deadweight Loss (DWL)

When markets are distorted by taxes, price controls or monopoly, total surplus falls. The reduction is the DEADWEIGHT LOSS — gains from trade that never happen.
Tariffs, minimum wages above equilibrium, rent controls all create DWL. Government intervention always involves this trade-off.

SECTION 02

Firm Costs & Revenue

Revenue Concepts — TR, AR, MR

Revenue concepts determine HOW MUCH a firm earns. They differ fundamentally between competitive and non-competitive markets.

TR = Price × Quantity AR = TR/Q = Price MR = ΔTR / ΔQ

Concept	Formula	Perfect Competition	Imperfect Competition
Total Revenue (TR)	P × Q	Rises proportionally with Q (P fixed)	Rises then falls: selling more requires lower P on ALL units
Average Revenue (AR)	TR / Q = P	Always equals price	Always equals price (same formula)
Marginal Revenue (MR)	d(TR)/dQ	MR = Price (horizontal demand curve)	MR < Price (must cut price to sell more)
MR relationship	MR vs P	MR = P — firm is a price TAKER	MR < P — firm has pricing POWER

Why MR < P under imperfect competition

To sell one more unit, a monopolist must lower price on ALL units sold. The extra revenue from the new unit is OFFSET by the price reduction on existing units.

If P falls from 60 to 59 to sell 101 instead of 100 units: TR rises by 59 (new unit) but falls by 100 (1 reduction on existing 100 units).

MR = 59 − 100 = −41. Dramatically below price.

Cost Concepts — TC, TFC, TVC, ATC, MC

$$TC = TFC + TVC \qquad ATC = TC/Q \qquad MC = \Delta TC/\Delta Q$$

Cost	Definition	Short Run	Long Run
Total Fixed Cost (TFC)	Costs that do NOT vary with output	Rent, depreciation, salaries of permanent staff	ALL costs are variable — no fixed costs exist
Total Variable Cost (TVC)	Costs that DO vary with output	Materials, hourly labour, utilities	All inputs adjustable
Average Total Cost (ATC)	TC per unit = TC/Q	U-shaped: falls then rises due to fixed cost spread and diminishing returns	U-shaped LRAC reflects scale (economies/diseconomies)
Marginal Cost (MC)	Cost of one extra unit of output	MC eventually rises due to diminishing marginal returns	Traces bottom of each short-run ATC

Economic vs Accounting Profit — a critical CFA distinction (LM1)

Accounting profit = TR − explicit costs (what appears on P&L).
Economic profit = TR − explicit costs − IMPLICIT costs (opportunity costs of owner's time, capital, etc.).

CFA LM1: 'Economic cost considers the total opportunity cost of all factors of production.'
A firm earning zero economic profit is earning a NORMAL return — exactly covering the opportunity cost of all inputs. This is actually fine — no incentive to exit. Positive economic profit → above-normal returns → attracts new entrants.

Profit Maximisation — The MR = MC Rule

Profit is maximised where MR = MC (and MC is not falling with output)

This is the UNIVERSAL profit-maximisation rule. It applies under all market structures.

If MR > MC	Each additional unit adds more to revenue than it costs. Expand production.
If MR < MC	Each additional unit adds more to costs than to revenue. Contract production.
If MR = MC	Profit is maximised. No incentive to change output in either direction.

Important: under perfect competition, MR = P, so profit maximisation means P = MC.
Under imperfect competition, P > MR = MC, so price always exceeds marginal cost.

CFA LM1: WR International — Breakeven & Profit Max

Directly from CFA Vol 2, LM1. Given a monopolist's TR and TC at various price levels, find breakeven points and profit maximum.

Price	Quantity	Total Revenue	Total Cost	Profit	Status
20	9,500	190,000	240,000	−50,000	Loss
30	9,250	277,500	300,000	−22,500	Loss
40	9,000	360,000	360,000	0	BREAKEVEN
50	8,750	437,500	420,000	17,500	Profit
60	8,500	510,000	480,000	30,000	MAX PROFIT
70	8,250	577,500	550,000	27,500	Profit
80	8,000	640,000	640,000	0	2nd BREAKEVEN
90	7,750	697,500	710,000	−12,500	Loss

CFA answers: (1) Breakeven at Q=40 where TR=TC=360,000. (2) Profitable from Q>40 to Q<80. (3) Max profit = 30,000 at Q=60.

Shutdown Decision — Short Run vs Long Run

Continue if TR ≥ TVC Shutdown if TR < TVC Exit if TR < TC (long run)

Revenue-Cost Relationship	Short-Run Decision	Long-Run Decision
TR = TC	Stay in market	Stay in market
TR = TVC but < TC	Stay in market (cover variable costs)	Exit market (cannot sustain fixed cost losses)
TR < TVC	SHUT DOWN production	Exit market

CFA LM1, Example 2 — London-Based Business Shutdown

A London business: Revenue = GBP 2m, TC = GBP 2.5m (TFC = GBP 1m, TVC = GBP 1.5m). Net loss = GBP 500,000.

Question 1 (short run): TR = 2m > TVC = 1.5m. Continuing minimises loss at 500k. If they shut down, they still owe TFC = 1m. CONTINUE.
Question 2 (long run): If revenue shortfall persists, exit market. Shareholders preserve equity by exiting early.
Question 3: Now TR = 1.3m, TC = 2.5m, loss = 1.2m. TR = 1.3m < TVC = 1.5m. SHUT DOWN NOW. Loss from shutdown = TFC = 1m < loss from operating = 1.2m.

Economies & Diseconomies of Scale (CFA LM1)

Economies of Scale = falling long-run ATC as output rises

Minimum Efficient Scale = bottom of LRAC

Economies of Scale

LRAC falls as output grows.

Causes:
- Specialisation: workers and machines focus on narrow tasks
- Indivisibility: large equipment spreads fixed cost over more units
- Buying power: bulk purchasing reduces input prices
- Financial advantages: large firms borrow at lower rates

Example: Amazon's logistics network gets cheaper per unit as volume grows.

Diseconomies of Scale

LRAC rises as output grows too large.

Causes:
- Bureaucratic inefficiency: communication and coordination costs
- Management span: too wide to monitor effectively
- Input price rises: bidding up scarce specialised labour
- Motivational problems: large organisations lose focus

Example: mega-mergers often fail to achieve projected cost savings.

Minimum Efficient Scale (MES): the output level at which LRAC is minimised. Firms must reach MES to be competitive. Below MES, new entrants or larger incumbents have a cost advantage.

Analyst note: industries with high MES (e.g. semiconductor fabs, aircraft) have high natural barriers to entry — supports incumbent pricing power and moat analysis.

Price Controls — Floors, Ceilings & Market Distortions

Governments sometimes override market equilibrium. These interventions always create trade-offs between equity and efficiency.

Price Floor (Minimum Price)

Set ABOVE equilibrium. Creates a SURPLUS.

Example: minimum wage above equilibrium wage → excess labour supply (unemployment).
Example: EU agricultural price supports → food surpluses.

Winners: sellers who keep their jobs/contracts at higher price.
Losers: buyers, and sellers excluded from market.

Price Ceiling (Maximum Price)

Set BELOW equilibrium. Creates a SHORTAGE.

Example: rent controls below market rent → housing shortage, long waiting lists.
Example: utility price caps → under-investment in new capacity.

Winners: buyers lucky enough to get the product at lower price.
Losers: sellers who exit market, buyers who cannot find product.

Deadweight Loss from Price Controls

Both price floors and ceilings create deadweight loss — gains from trade that do not happen.
Rent control: landlords supply fewer units (not worth the regulated rent) + tenants cannot find apartments.
Minimum wage: some workers priced out of jobs. Net welfare effect is negative even if some benefit.

Long-Run Average Cost (LRAC) — The Planning Horizon

In the long run, all costs are variable. The LRAC curve is the ENVELOPE of many short-run ATC curves, each for a different plant size.

LRAC traces the LOWEST possible ATC for each output level when ALL inputs are optimised

Economies of scale	LRAC falls as output increases. Each doubling of output costs less than double. Specialisation, indivisibilities and bulk purchasing drive this. Relevant at low output levels.
Constant returns to scale	LRAC is flat. Doubling inputs doubles output at the same cost per unit. The 'sweet spot' of efficient operation. May extend over a range of outputs.
Diseconomies of scale	LRAC rises as output grows too large. Bureaucracy, co-ordination failures and input price pressure dominate. Mega-mergers often enter this zone.

Minimum efficient scale (MES) = bottom of LRAC. Determines minimum viable firm size. High MES = high natural barriers to entry.

SECTION 03

Market Structures (CFA LM1)

Market Structures Overview — CFA LM1, Exhibit 13

CFA Vol 2 LM1 classifies all markets into four structures based on five characteristics. This is the core analytical framework.

Market Structure	No. Sellers	Product	Barriers	Pricing Power	Non-price Comp
Perfect Competition	Many	Homogeneous	Very Low	None	None
Monopolistic Competition	Many	Differentiated	Low	Some	Advertising, differen
Oligopoly	Few	Homogeneous/Diff.	High	Some/Considerable	Advertising, differen
Monopoly	One	Unique	Very High	Considerable	Some advertising

CFA LM1 Analyst Insight

From investor perspective: most desirable to OWN = monopoly or oligopoly (pricing power, high margins). Most desirable as CONSUMER = perfect competition (lowest prices). Financial analysts must understand market structure to forecast a firm's future profit stream. Pricing power = sustainable economic profit.

A firm's market structure determines its MOAT — the ability to protect economic profit from competition. This is central to equity analysis.

Perfect Competition — Price Takers, Zero Long-Run Economic Profit

In perfect competition, firms are PRICE TAKERS. No individual firm can influence the market price.

Characteristics

- Many small firms, none dominant
- Homogeneous (identical) products
- Perfect information for all buyers/sellers
- Free entry and exit — no barriers
- Firm's demand curve is HORIZONTAL at market price
- MR = P = AR at all output levels

Long-Run Equilibrium

- Positive economic profit attracts new entrants
- New entrants shift supply right, price falls
- Process continues until P = ATC (zero economic profit)
- Negative profit: firms exit, supply falls, price rises back
- Long-run result: P = MR = MC = ATC (minimum ATC)
- Investors cannot expect sustained above-normal returns

Perfect competition examples: commodity farmers (wheat, corn, soybeans), foreign exchange markets, some financial markets.

Monopolistic Competition — Coca-Cola and the Power of Differentiation

Many firms, differentiated products. Firms compete on brand, quality, advertising — not just price. CFA LM1 uses Coca-Cola as its example.

CFA LM1 EXAMPLE — Coca-Cola Monopolistic Competition

CFA states: 'If the seller can convince consumers that its product is uniquely different from other similar products, then the seller can exercise some degree of pricing power.' 'Many of Coca-Cola's customers believe that their beverages are truly different from and better than all other soft drinks.'

Same is true for fashion creations and cosmetics. The brand loyalty = perceived differentiation = pricing power.
Other examples: restaurants, hairdressers, clothing retailers, legal services, petrol stations on motorway junctions.

Short-Run: Positive Profit Possible

Product differentiation allows above-normal returns short-term. New entrants cannot instantly copy brand identity or location. Advertising spending to reinforce differentiation.

Long-Run: Zero Economic Profit

Low barriers to entry attract imitators. Substitutes proliferate. Demand for original firm falls. Long-run: P = ATC but P > MC (unlike perfect competition). Some excess capacity remains.

Oligopoly — Strategic Interdependence & OPEC

Few firms. Each firm's pricing decision depends on ANTICIPATED REACTIONS of rivals. Strategic behaviour dominates. CFA uses commercial airlines and crude oil as examples.

CFA LM1 — OPEC Crude Oil Cartel Example

CFA states: 'In 2016, the average extraction cost of a barrel of crude oil from Saudi Arabia was approximately $9, while the average cost from US shale was roughly $23.50.'

The similarity of Gulf producers' cost structures (~$9-10/barrel) makes collusion more stable. If OPEC had kept crude below $30, there would have been no economic case to develop US shale fields through fracking. OPEC's cartel raised prices to a level where outside sources became economic — increasing competition the cartel faces.

Lesson: successful collusion requires similar cost structures and few firms. Rising prices attract new entrants who undermine the cartel.

Six factors affecting collusion success (CFA LM1): number of firms, product similarity, cost structure similarity, order size/frequency, threat of retaliation, degree of external competition.

Kinked Demand Curve Theory

If one firm raises price, rivals do NOT follow → loses market share. If one firm cuts price, rivals DO follow to protect share. Result: prices tend to be STICKY in oligopoly markets.

Barriers to Entry

Oligopoly requires high barriers: large capital requirements, patents, network effects, regulatory licences. Example: global credit card networks (Visa/Mastercard), semiconductor equipment.

Monopoly — One Seller, Maximum Pricing Power

A pure monopoly is the single supplier of a good with no close substitutes. The monopolist IS the market demand curve — it chooses P and Q subject to demand.

Monopolist maximises profit at MR = MC Sets price P > MR = MC (positive price-cost margin)

Feature	Monopoly vs Perfect Competition	Analyst Implication
Price	P > MC (monopoly) vs P = MC (competitive)	Higher price → higher revenue per unit
Quantity	Q < competitive equilibrium	Less produced → artificial scarcity
Economic profit	Can sustain long run (barriers block entry)	Investable: high margins, strong cash flow
Consumer surplus	Lower (more transferred to producer)	Allocative inefficiency → DWL
Regulation	Often regulated by government	Price set to allow normal return (RAB models, utilities)

CFA: 'Investors may benefit from being shareholders of monopolistic firms that have large margins and substantial positive cash flows.'

SECTION 04

Game Theory & Market Power (CFA LM1)

Nash Equilibrium — ArcCo/BatCo Duopoly (CFA LM1)

Directly from CFA Vol 2, LM1. The Nash equilibrium is where no firm can improve profits by unilaterally changing its strategy.

PAYOFF MATRIX: (ArcCo profit, BatCo profit)

	ArcCo: LOW PRICE	ArcCo: HIGH PRICE
BatCo: LOW	(50, 70)	**(300, 350)**
BatCo: HIGH	(80, 0)	(500, 300)

CFA LM1: Finding the Nash Equilibrium

For BatCo: If ArcCo plays Low, BatCo prefers Low (70>0). If ArcCo plays High, BatCo still prefers Low (350>300). BatCo's DOMINANT strategy = Low.
For ArcCo: if BatCo plays Low, ArcCo prefers High (300>50). If BatCo plays High, ArcCo prefers High (500>80). ArcCo's DOMINANT strategy = High.
NASH EQUILIBRIUM = ArcCo High, BatCo Low → profits = (300, 350). Highlighted above.
Joint maximum would be (500,300)=800 if both chose High, but BatCo deviates to Low to earn 350>300. Collusion is tempting but

Concentration Ratio — CFA LM1 Question Set

The concentration ratio = sum of market shares of the largest N firms. Simplest measure of market power. From CFA Vol 2, LM1 Question Set:

CFA LM1 QUESTION SET — 6-Firm Industry

Company Sales (EUR millions): ABC=300, Brown=250, Coral=200, Delta=150, Erie=100, All others=50
Total industry sales = 300+250+200+150+100+50 = EUR 1,050m

4-firm concentration ratio = (300+250+200+150) / 1,050 = 900/1,050 = 86% (CFA answer B)

Interpretation: top 4 firms control 86% of the market. Highly concentrated — suggests oligopolistic conditions.

Limitation: if the 2 largest firms merge (300+250=550), the top 4 would now be: 550+200+150 = 900 still = 86%. The CR does NOT detect this consolidation of power. This is why the HHI is preferred by regulators.

Concentration Ratio Formula

CR-N = Σ(market shares of top N firms)
Range: 0% (perfect comp) to 100% (monopoly)
CR-4 > 80% = typically oligopolistic
Easy to compute but ignores distribution within top N

Key Limitation

Insensitive to mergers among top incumbents.
Cannot detect difference between:

- 1 firm with 60% + 2 firms with 10% each
- 3 firms each with 30% each

Same CR, very different competitive dynamics.

Herfindahl-Hirschman Index (HHI) — CFA LM1 Practice Q3

$HHI = \Sigma(\text{market share}_i)^2$ Range: 1/n (equal shares) to 1.0 (monopoly)

Directly from CFA Vol 2, LM1 Practice Problems Q3. HHI solves the merger detection problem of the CR.

CFA LM1 PRACTICE Q3 — 5-Firm Market: Zeta=35%, Yusef=25%, Xenon=20%, Waters=10%, Vlastos=10%

3-firm HHI = $0.35^2 + 0.25^2 + 0.20^2 = 0.1225 + 0.0625 + 0.0400 = 0.225$ (CFA answer B)

Now suppose Zeta (35%) and Yusef (25%) MERGE to form one firm with 60%:
Post-merger 3-firm HHI = $0.60^2 + 0.20^2 + 0.10^2 = 0.36 + 0.04 + 0.01 = 0.410$

HHI nearly DOUBLES (0.225 → 0.410) — captures the concentration increase.
Compare: CR-3 was 80% pre-merger (35+25+20), becomes 90% post-merger (60+20+10) — only a 10pp change.

US antitrust guidelines: HHI > 0.25 = highly concentrated. Mergers raising HHI by > 0.025 warrant scrutiny.

CFA LM1 Example 4 (10 equal firms of 10%): CR-4 = 40%, HHI = $0.10^2 \times 4 = 0.04$ (CFA answer C).

Market Power — Investment Implications for Analysts

Understanding market structure is not just economics theory — it directly drives equity valuation and credit analysis.

Structure	Economic Profit (LR)	Key Investment Signal	Example Sectors
Perfect competition	Zero	Price taker. Margin competition. Hard to own at premium.	Agricultural commodities, generic drugs
Monopolistic competition	Near zero LR	Depends on brand strength. Look for pricing power durability.	Restaurants, retail, beauty brands
Oligopoly	Positive (if stable)	High margins if collusive. Antitrust risk. Watch barriers to entry.	Airlines, telecom, cement, ratings agencies
Monopoly	Positive (regulated)	Regulated return. RAB model. Stable, bond-like cash flows.	Water, gas distribution, power grids

The Moat Framework

Warren Buffett's 'economic moat' is essentially the CFA market structure framework in investor language:

- Wide moat = monopoly or strong oligopoly (pricing power, barriers to entry, sustainable economic profit)
- Narrow moat = monopolistic competition with real brand advantage
- No moat = perfect competition (commodity-like, price taker)

Analysts price moats using premium P/E, EV/EBITDA multiples and DCF terminal value assumptions.

Oligopoly Pricing Strategies — Kinked Demand Curve

The kinked demand curve model explains why prices tend to be STICKY in oligopolies — firms fear asymmetric rival reactions.

If I raise price: rivals do NOT follow → I lose share. If I cut price: rivals DO follow → no net gain.

The Logic of Price Stickiness in Oligopoly

Demand above current price is ELASTIC (rivals do not match — customers switch to them): raising price loses significant volume.
Demand below current price is INELASTIC (rivals match any cut to protect their share): cutting price gains little extra volume.

Result: there is a KINK at the current price. The MR curve is DISCONTINUOUS at this output level.
As long as MC passes through the gap in the MR curve, the profit-maximising price does NOT change even when costs change moderately.

This explains the empirical observation that oligopoly prices remain stable over long periods despite cost fluctuations.

Cournot Model

Firms choose QUANTITY simultaneously, assuming rivals hold quantity fixed. Output between competitive (too high) and monopoly (too low). Price competition softens as more firms enter.

Stackelberg Model

Leader sets quantity FIRST; follower observes and reacts. First-mover advantage: leader captures larger market share. Example: established airline adding routes before rivals.

Price Discrimination — Extracting Consumer Surplus

Price discrimination occurs when a firm with pricing power charges different prices to different customers for the SAME product.

Degree	How It Works	Example	Condition Required
1st Degree (perfect)	Each buyer pays their maximum willingness to pay. Producer captures ALL consumer surplus.	Negotiated deals, bespoke consulting	Perfect knowledge of each buyer's WTP
2nd Degree	Price varies by quantity consumed. Bulk discounts or tiered pricing.	Utility tariffs, broadband data bundles, airline classes	Observable purchase behaviour
3rd Degree	Different prices to different GROUP segments based on observable characteristics.	Student discounts, senior fares, peak/off-peak rail, geography-based pricing	Ability to prevent resale between groups

When is price discrimination profitable?

Three conditions required: (1) the firm has pricing power (not a price taker), (2) customers have different elasticities of demand, (3) resale between segments can be prevented.

Profit impact: price discrimination INCREASES total profit vs single price. Higher-price segments subsidise access for lower segments. Often total output is HIGHER than under single pricing, reducing deadweight loss.

Examples: airline revenue management, SaaS pricing tiers, pharma pricing by country.

Formula Reference Card — Session 3 Microeconomics

All key formulas and concepts from this session.

Price Elasticity of Demand	PED = % ΔQ^d / % ΔP |PED|>1 elastic, <1 inelastic, =1 unit elastic
Income Elasticity	YED = % ΔQ^d / % ΔIncome >0 normal good, <0 inferior good, >1 luxury
Cross-Price Elasticity	XED = % ΔQ_a / % ΔP^b >0 substitutes, <0 complements
Total Revenue	TR = P × Q
Marginal Revenue	MR = ΔTR / ΔQ (= P in perfect competition, < P otherwise)
Profit Maximisation	MR = MC (and MC not falling). Economic profit = TR − TC (incl. opportunity costs)
Breakeven	TR = TC, i.e. P = ATC
Shutdown rule	Shut down if TR < TVC in short run. Exit if TR < TC in long run.
Consumer Surplus	Area between demand curve and price paid (triangle)
Producer Surplus	Area between price received and supply curve (triangle)
Concentration Ratio	CR-N = sum of top N market shares. Range 0%–100%.
HHI	HHI = $\Sigma(\text{market share}_i)^2$. Range 1/N (equal shares) to 1.0 (monopoly). HHI>0.25 = highly concentrated.

PRACTICE PROBLEMS

Work through each before checking answers.

Q1	WR International (CFA LM1): At price=60, quantity=8,500, TR=510,000, TC=480,000. Is this profit-maximising?
Q2	London business (CFA LM1): TR=GBP1.3m, TVC=GBP1.5m, TFC=GBP1m. Should the firm operate in the short run? What is the loss from continuing vs shutdown?
Q3	HHI (CFA LM1): Industry has 5 firms: Zeta=35%, Yusef=25%, Xenon=20%, Waters=10%, Vlastos=10%. What is the 3-firm HHI? If Zeta and Yusef merge, what is the new HHI?
Q4	A firm has a PED of 0.4 and currently charges GBP10. If it raises price by 20%, what happens to quantity demanded and total revenue? Is this rational?
Q5	ArcCo/BatCo (CFA LM1 Nash): In the payoff matrix, both firms could earn joint profit of 800 if both price High. Why does this not happen in the Nash equilibrium?

Practice — Worked Solutions

A1

At Q=8,500 (price=60): profit=30,000 (max). At Q=8,750 (price=50): profit=17,500. Reducing price to sell more REDUCES profit. The firm is already at the maximum. MR<MC beyond Q=60 units: expanding costs more than it earns. This confirms Q=60 (price=60) is the profit-maximising output per CFA.

A2

TR=1.3m < TVC=1.5m. The firm CANNOT cover variable costs. Shutdown rule triggered. Loss from continuing = 1.2m. Loss from shutdown = TFC = 1.0m. Shut down saves GBP200,000. CFA Example 2 conclusion: shut down production. Shareholders save GBP200,000 in equity value.

A3

3-firm HHI = $0.35^2+0.25^2+0.20^2$ = 0.1225+0.0625+0.04 = 0.225 (CFA answer B). Post-merger: Zeta+Yusef=60%. HHI = $0.60^2+0.20^2+0.10^2$ = 0.36+0.04+0.01 = 0.410. HHI nearly doubles — regulators would scrutinise this merger. CR-3 only rises from 80% to 90% — less informative.

A4

PED=0.4 (inelastic). Price rises 20%. %ΔQ = −0.4×20% = −8%. Quantity falls 8%. New TR: if original P=10 and Q=100: new P=12, new Q=92. Old TR=1,000. New TR=12×92=1,104. TR RISES by 10.4%. Rational: inelastic demand means raising price increases total revenue. Standard pricing power analysis.

A5

ArcCo's dominant strategy is High (300>50 if BatCo Low; 500>80 if BatCo High). BatCo's dominant strategy is Low (70>0 if ArcCo Low; 350>300 if ArcCo High). Nash = ArcCo High, BatCo Low = (300, 350). Neither wants to deviate unilaterally. To reach (500,300), both must commit to High simultaneously — this requires collusion, which is typically illegal and unstable (BatCo would defect to earn 350>300).

KEY TAKEAWAYS

Session 3 — Microeconomics

- CFA WR International: profit max at Q=60 (price=60), profit=30,000. Breakeven at Q=40 and Q=80.
- London business (CFA): TR=2m > TVC=1.5m → continue short-run (save 500k vs 1m TFC). TR=1.3m < TVC=1.5m → shut down.
- PED: |PED|>1 elastic, cut price → TR rises. |PED|<1 inelastic, raise price → TR rises. Pricing power = inelastic demand.
- CFA LM1 Exhibit 13: 4 market structures. Only perfect competition has MR=P. Others: MR<P. Monopoly/oligopoly sustain economic profit.
- Nash equilibrium ArcCo/BatCo (CFA): ArcCo High/BatCo Low = (300,350). Collusion (both High=800 joint) is tempting but unstable.
- 4-firm CR = 86% (CFA LM1). 3-firm HHI = 0.225 pre-merger → 0.410 post-merger. HHI better than CR for detecting consolidation.